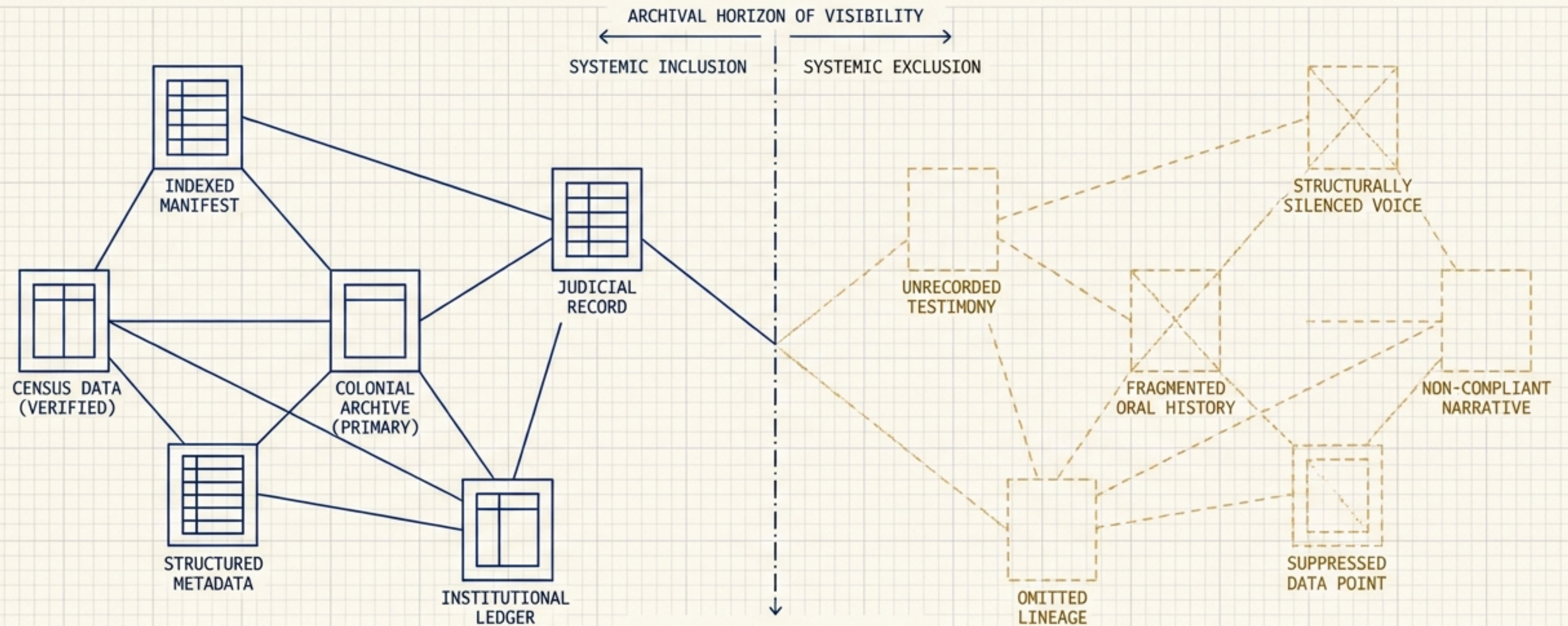


Engineering Epistemic Justice

Operationalizing History from Below through Structured Data

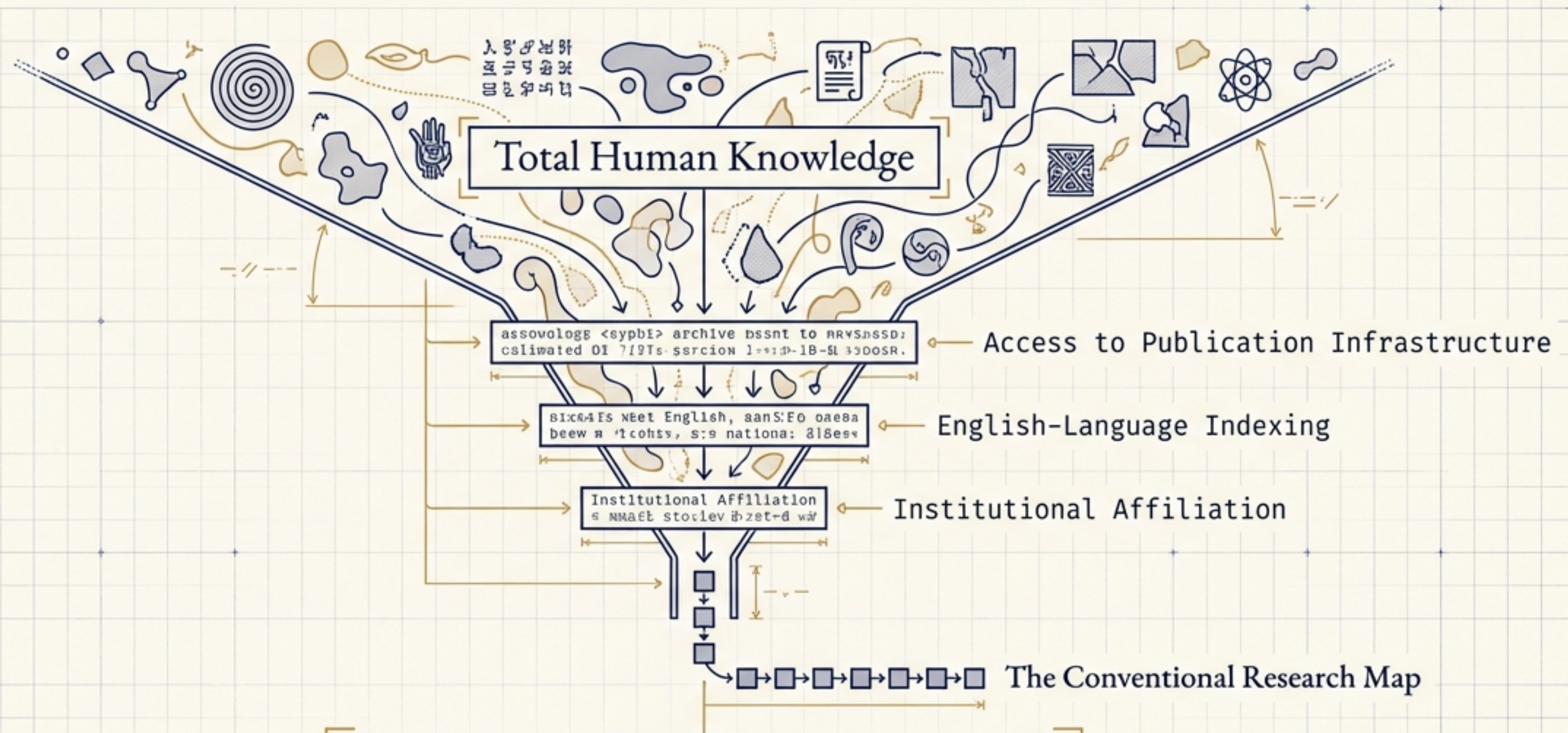


■ SOLID INDIGO: VISIBLE / INDEXED SOURCE

□ DASHED OCHRE: UNSEEN / SILENCED VOICE

The Map Only Shows What the Database Indexes

Research maps inherit the biases of whoever builds them. A literature review conducted through database search overwhelmingly finds institutional, English-language, text-based sources.



The Philosophy



History from Below
(E.P. Thompson)



Archival Silence
(Michel-Rolph Trouillot)



Standpoint Epistemology
(Sandra Harding)

The Code

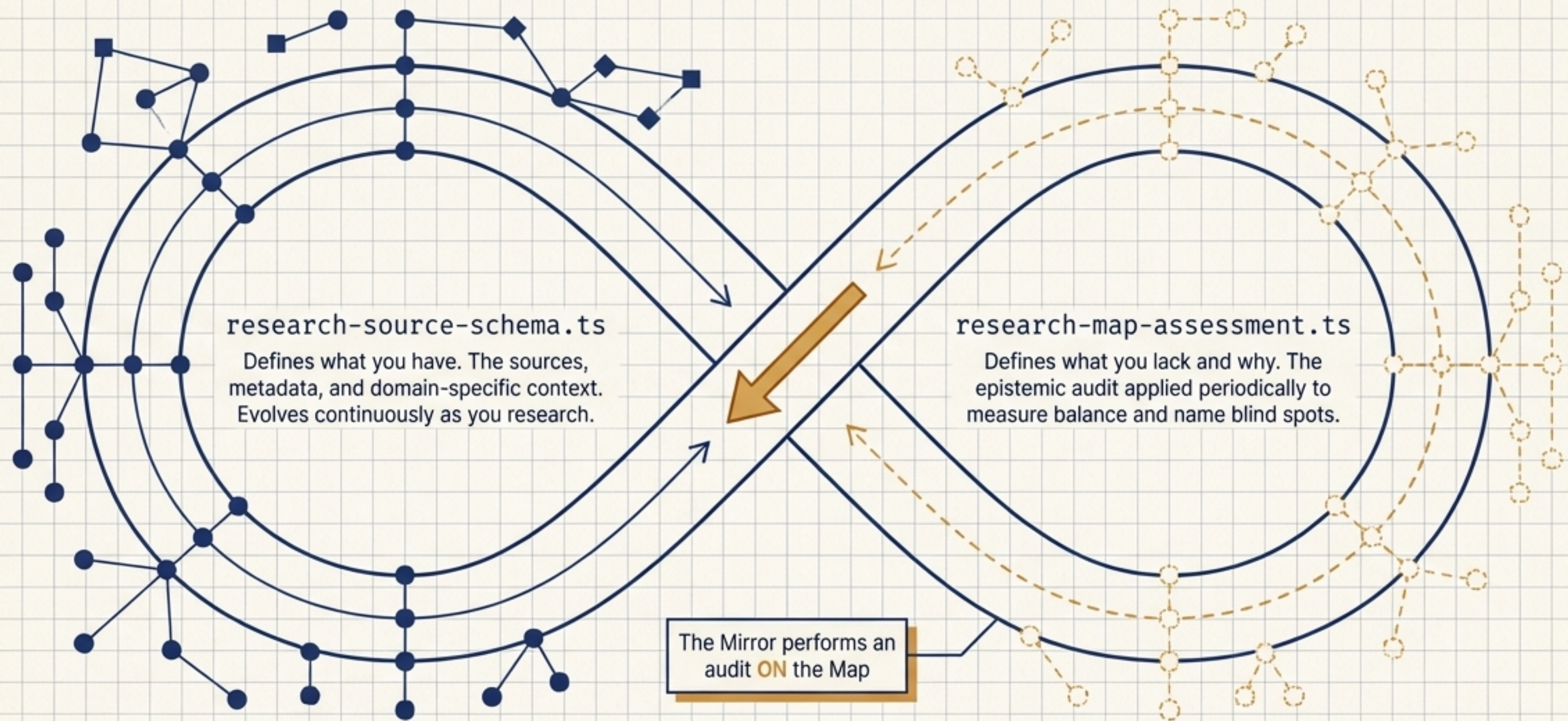
```
const MapAssessment = z.object({...}) {..}
```

```
structurallySilenced: z.array(z.string()) {..}
```

```
positionality: PositionalityStatement {..}
```

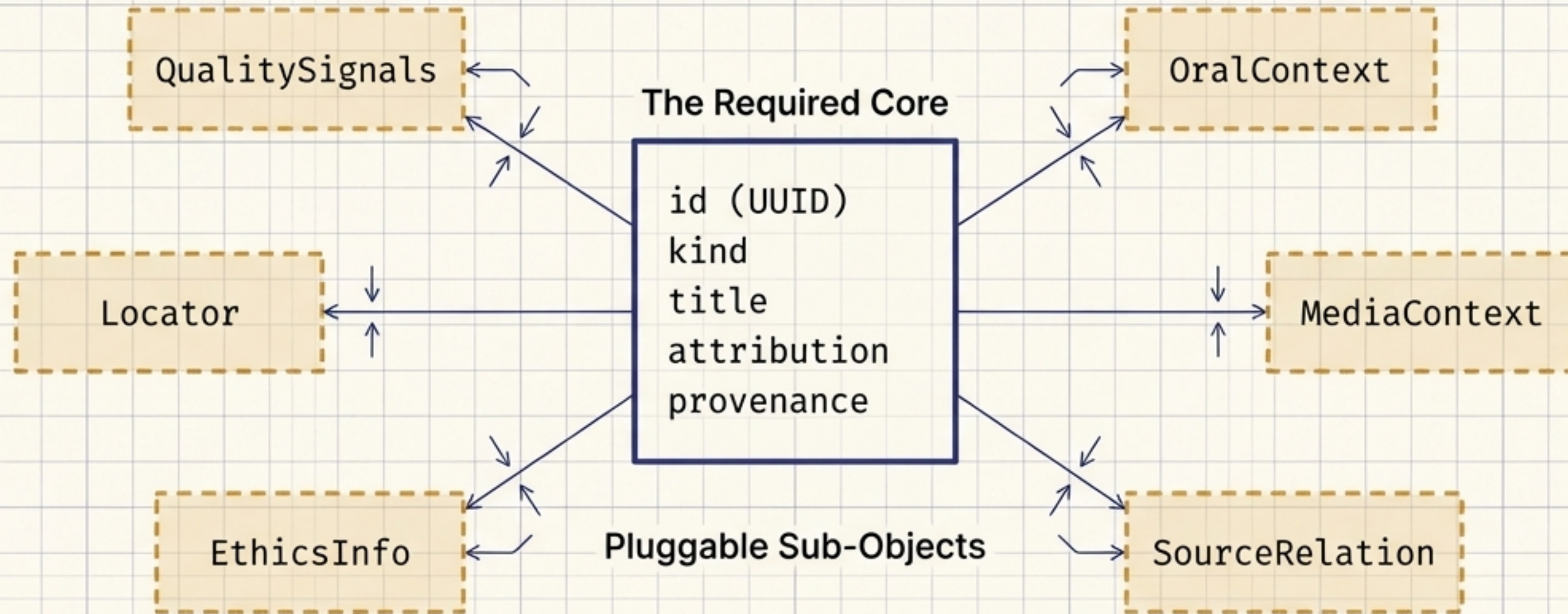
Most tools treat bias as someone else's problem. These schemas treat it as a structural property of the map itself—something that can be measured, named, and acted on.

The Map and The Mirror



Sparse by Default. Infinite in Scope.

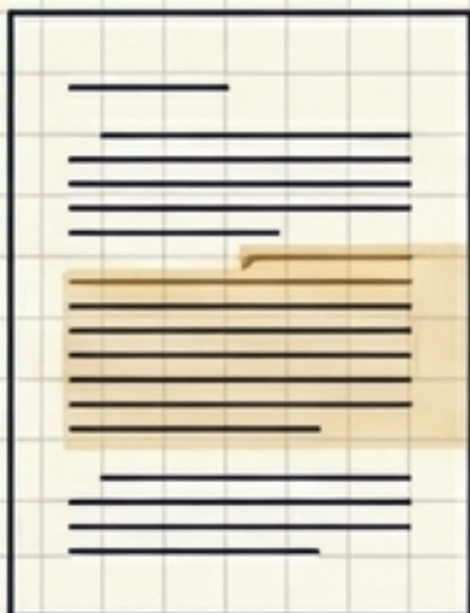
By keeping the core sparse, the system refuses to force false completeness. A fragmented oral history and a fully indexed DOI-backed journal article utilize the exact same TypeScript type.



Rebuilding the Standards of Evidence

Dimension	Conventional Standard	Research Map Schema
Temporal	ISO 8601 (Brittle, forces false precision)	FuzzyDate (Allows bounding notBefore/notAfter + confidence scores. "Sometime in the 1940s" is honest; "1940-01-01" is a lie)
Authorship	Individual Author (Western bias)	CommunalAttribution (Acknowledges oral/folk ownership, collective studios, and anonymity for safety)
Access & Ethics	Open vs. Closed	EthicsInfo (Captures living community consent, posthumous restrictions, and cultural sensitivities)

Pointing Precision, Regardless of Medium



LocatorPage

```
{ type: 'page',  
  value: '42-45'  
}
```

LocatorTime



```
{ type: 'time', startSeconds: 5004, endSeconds: 5050 }
```

LocatorSpatial



```
{ type: 'spatial', x: 0.1, y: 0.6, width: 0.2, height: 0.3 }
```

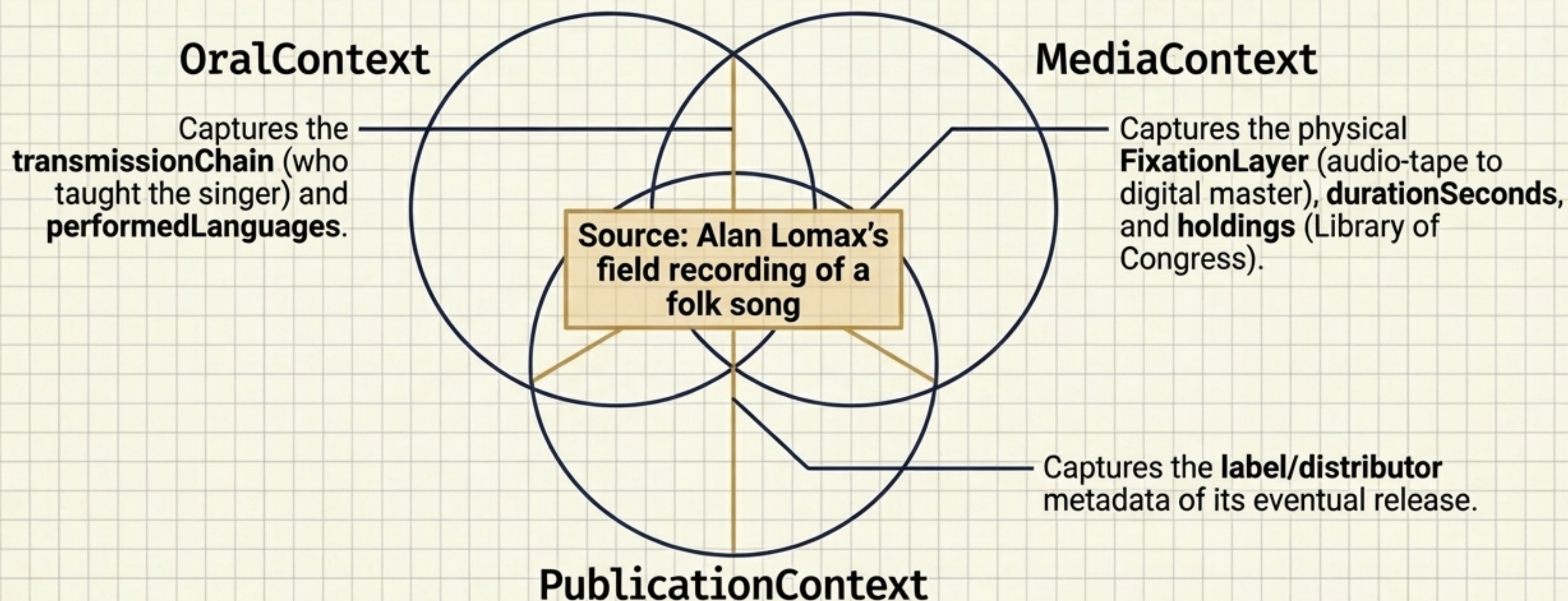
LocatorSection



```
{ type: 'section', value: 'Movement III' }
```


Triangulating Multi-Modal Context

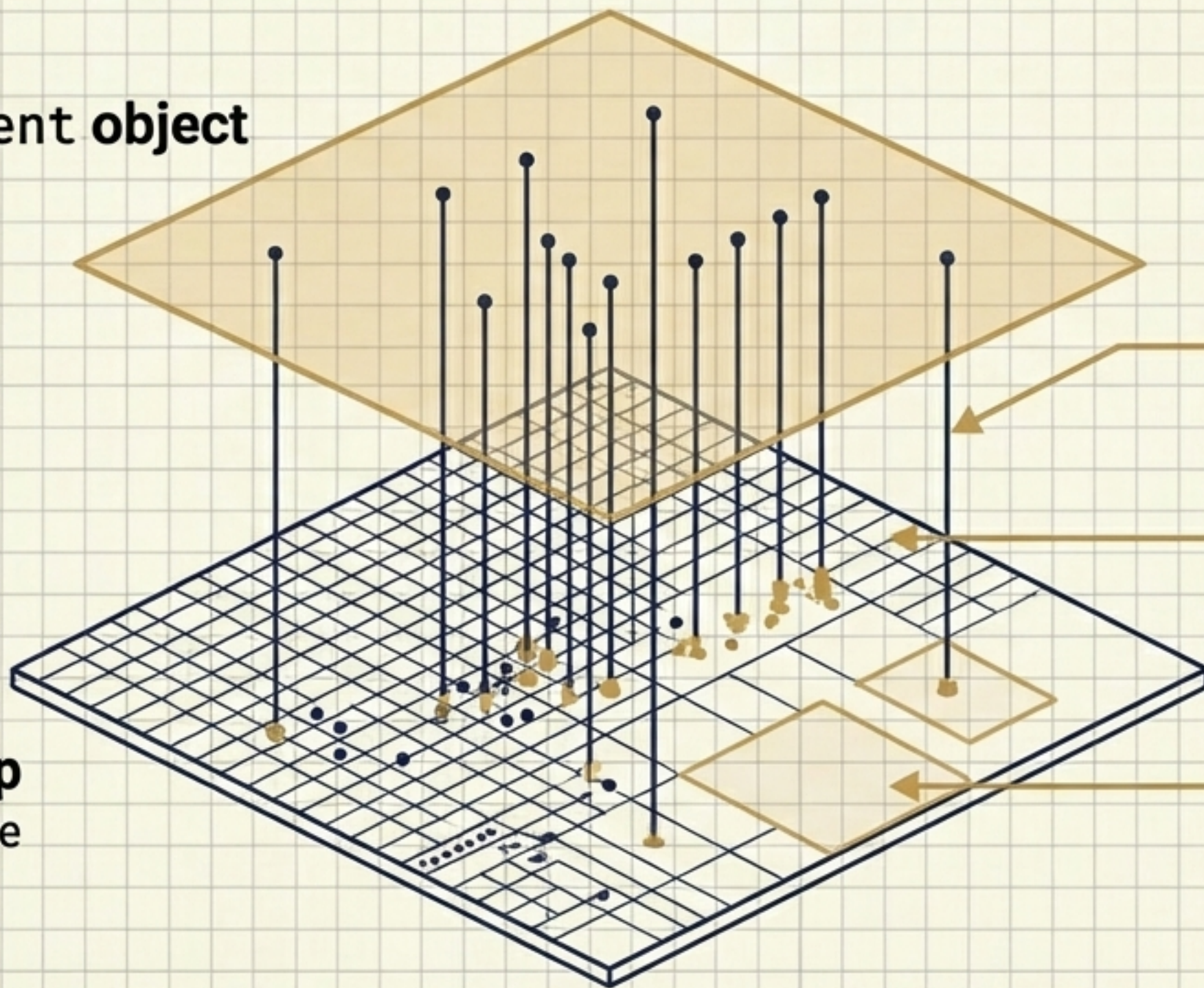
A single UUID simultaneously tracks the human lineage, the physical medium, and the commercial distribution without data collision.



The Mirror: Assessing the Voids

MapAssessment **object**

Source Map
Fira Code



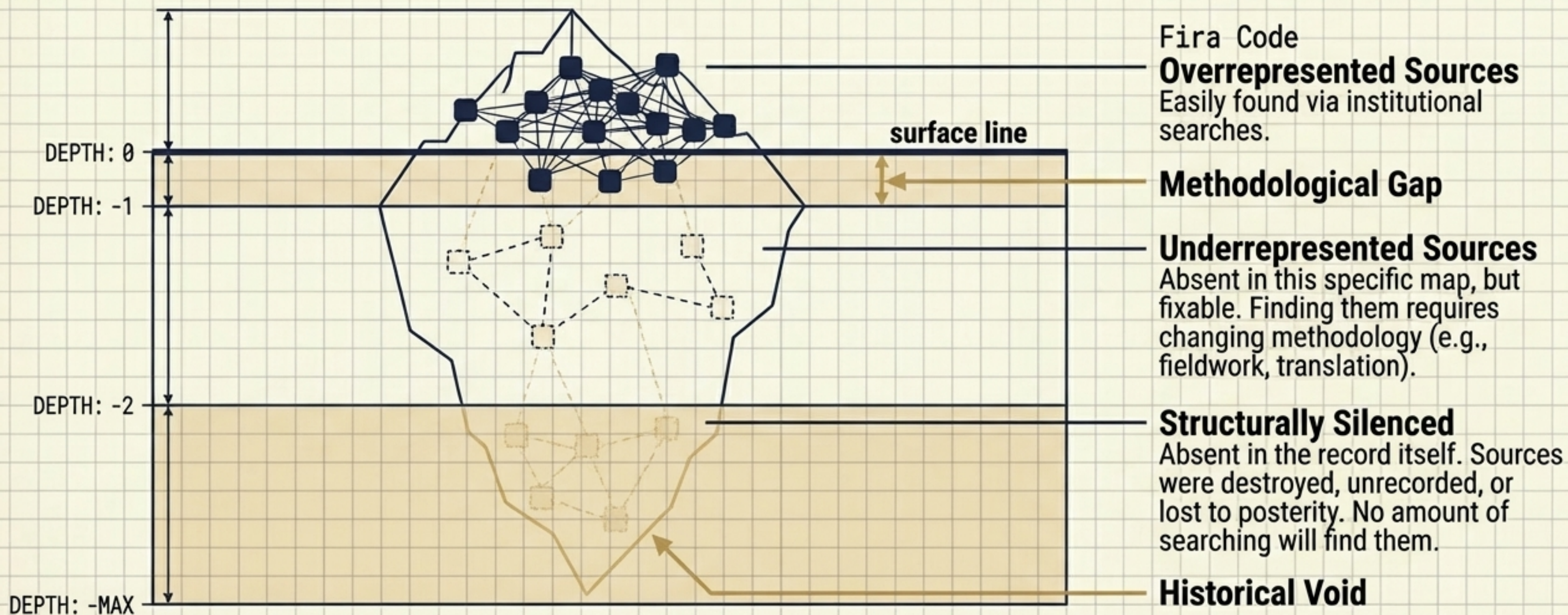
1. Power Axes: Uneven distribution across class, race, language, and geography.

2. Methodological Biases: What your retrieval methods (e.g., database search) systematically miss.

3. Blind Spots: Named, UUID-backed voids in the research coverage.

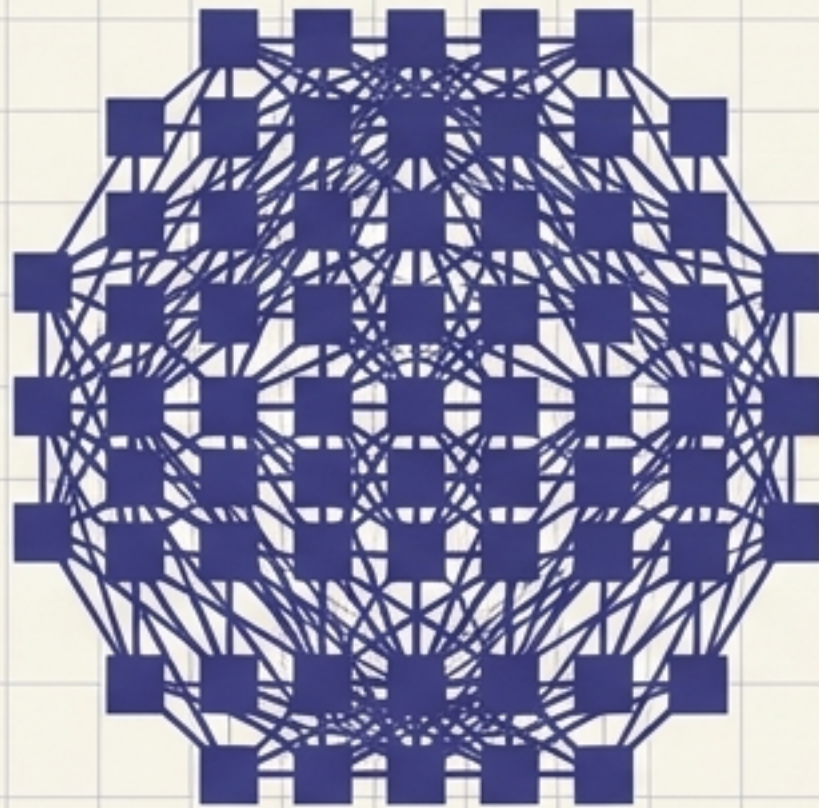
Not All Silences Are Created Equal

The schema forces the researcher to explicitly declare which type of absence they are facing.



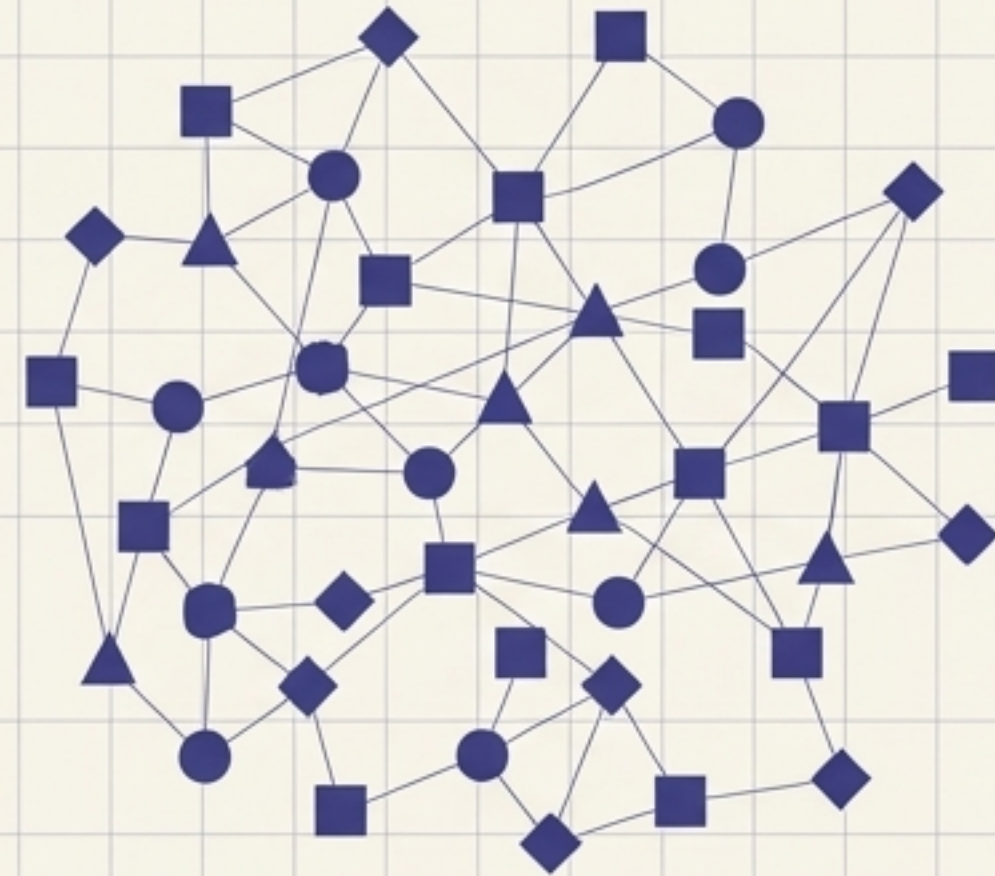
The Danger of Source Monocultures

Rigor is not just source count. A map of 1,000 journal articles and 0 oral testimonies systematically excludes everyone who doesn't publish in journals.



High Concentration

Map of 87% Journal Articles.
Herfindahl-Hirschman Index (HHI) ~ 0.76 .
High HHI means the map can only see what academic publishing can see.

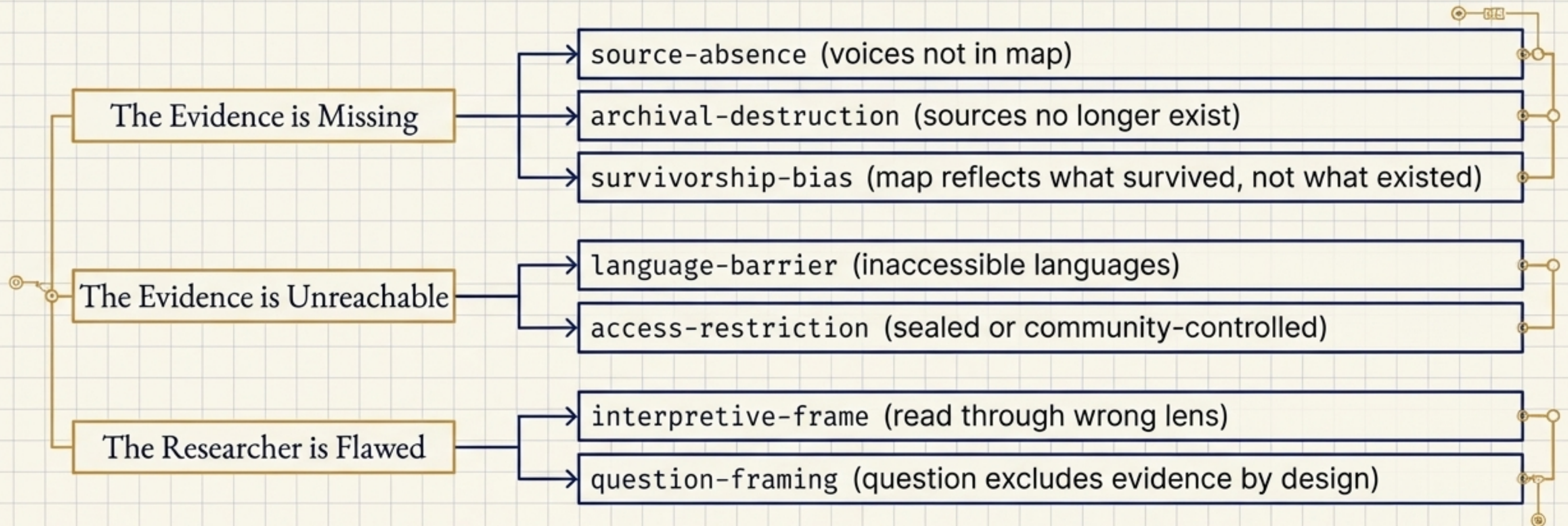


Low Concentration

Diverse ecosystem of oral histories, visual arts, archival documents, and field recordings. Low HHI metric.

Blind Spots as First-Class Objects

The schema elevates these from methodological footnotes to formal, addressable data structures.



SYSTEM ANNOTATION

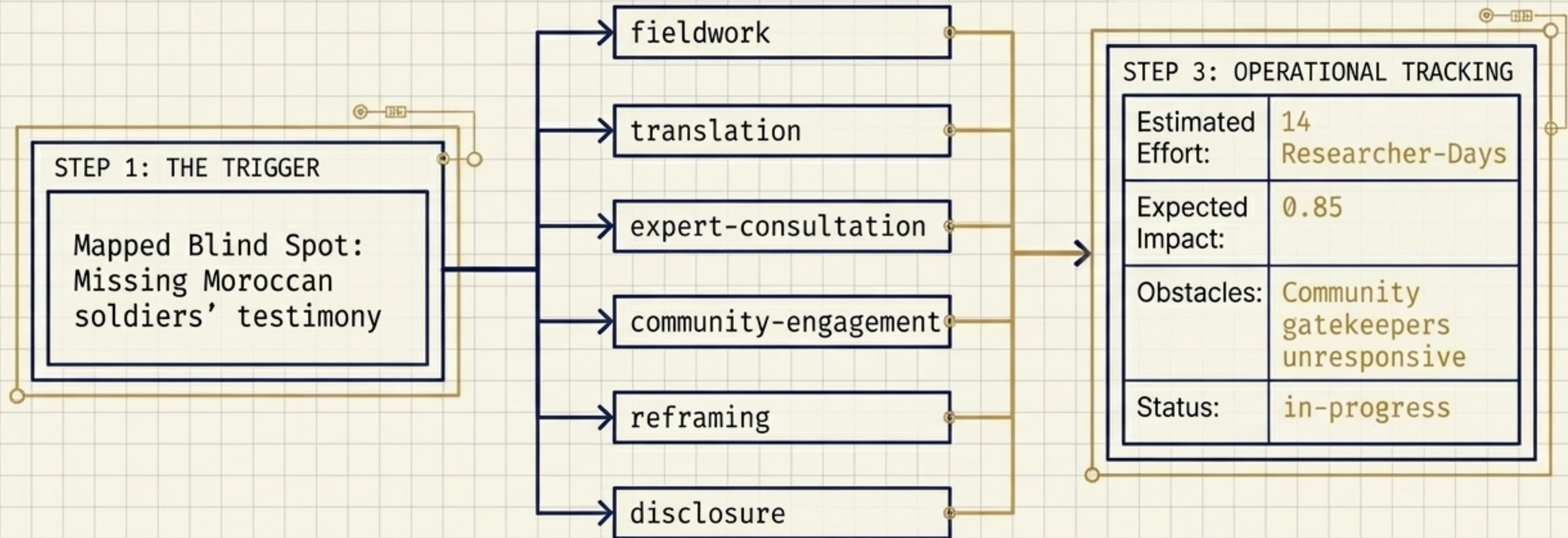
FIRA CODE

These are not methodology footnotes. They are
UUID-backed objects, severity-scored, and
programmatically linked to the specific claims they weaken.

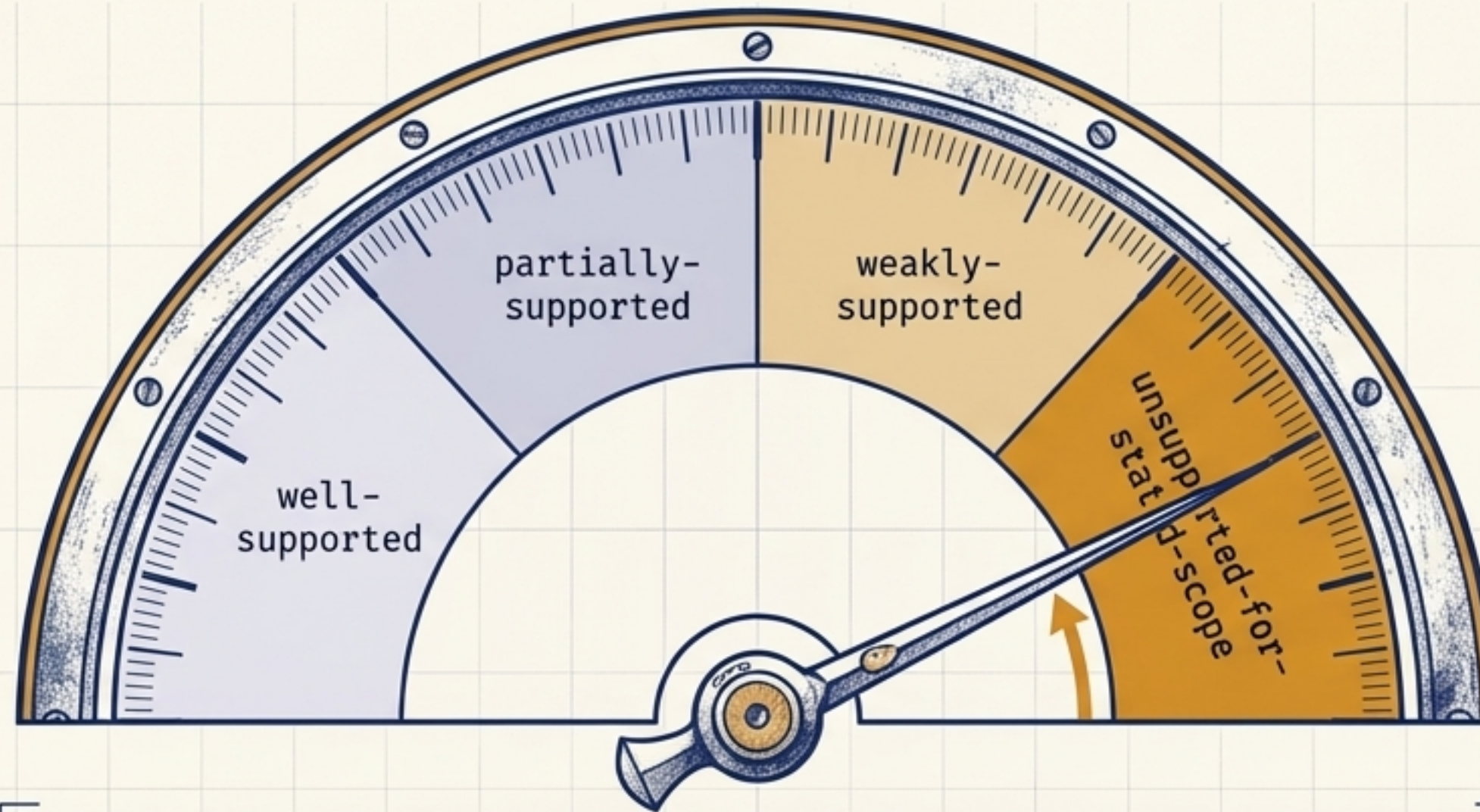


Performative Epistemology vs. Actionable Engineering

"Noting a blind spot without proposing an action is performative. The schema enforces operational follow-through."



Does the Evidence Base Support the Scope?



SCHEMA RULE ENFORCEMENT

The Thompson Test: You cannot claim to study "the working class" or "civilian experience" based solely on parliamentary debates and what elites wrote about them. If your sources do not match your scope, the schema forces a failing verdict.

SYSTEM VERDICT: SCOPE_FAILURE

The View From Somewhere

```
const PositionalityStatement = z.object({  
  background: NonEmpty,  
  languages: z.array(z.string()),  
  institutionalAccess: z.array(z.string()),  
  knownLimitations: z.array(z.string()),  
  motivation: z.string()  
});
```

Schema Limitations (Self-Critique)

- **No automated computation:** It is a data structure, not an analysis engine.
- **English-centric taxonomy:** Documentation and enum labels bias English.
- **Western-framework bias:** Default power axes privilege Western social categorizations.

Knowledge is always situated. The schema exists to prevent the researcher from hiding behind false neutrality.